

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Perform FFT and IFFT on discrete time signal.	
Experiment No: 5	Date:02-10-2026	Enrollment No:92400133030

GitHub Link:-

Code:- import numpy as np

import matplotlib.pyplot as plt

signal = np.array([1,2,3,4,5,6,7,8])

FFT

fft = np.fft.fft(signal)

Real and Imaginary

real = np.real(fft)

imag = np.imag(fft)

Magnitude and Phase

magnitude = np.abs(fft)

phase = np.angle(fft)

Reconstructed Signal

reconstructed = np.real(np.fft.ifft(fft))

4 Plots

plt.figure(figsize=(8,8))

plt.subplot(4,1,1)

plt.stem(real, label="Real")

for i, val in enumerate(real):

plt.annotate(f'{val:.2f}', (i, val), textcoords="offset points", xytext=(0,10), ha='center',
fontSize=8)

plt.stem(imag, label="Imaginary")

for i, val in enumerate(imag):

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```
plt.annotate(f'{val:.2f}', (i, val), textcoords="offset points", xytext=(0,-15), ha='center',
fontSize=8)
```

```
plt.title("FFT Real and Imaginary")
```

```
plt.legend()
```

```
plt.subplot(4,1,2)
```

```
plt.stem(magnitude)
```

```
for i, val in enumerate(magnitude):
```

```
plt.annotate(f'{val:.2f}', (i, val), textcoords="offset points", xytext=(0,10), ha='center',
fontSize=8)
```

```
plt.title("Magnitude")
```

```
plt.subplot(4,1,3)
```

```
plt.stem(phase)
```

```
for i, val in enumerate(phase):
```

```
plt.annotate(f'{val:.2f}', (i, val), textcoords="offset points", xytext=(0,10), ha='center',
fontSize=8)
```

```
plt.title("Phase")
```

```
plt.subplot(4,1,4)
```

```
plt.stem(reconstructed)
```

```
for i, val in enumerate(reconstructed):
```

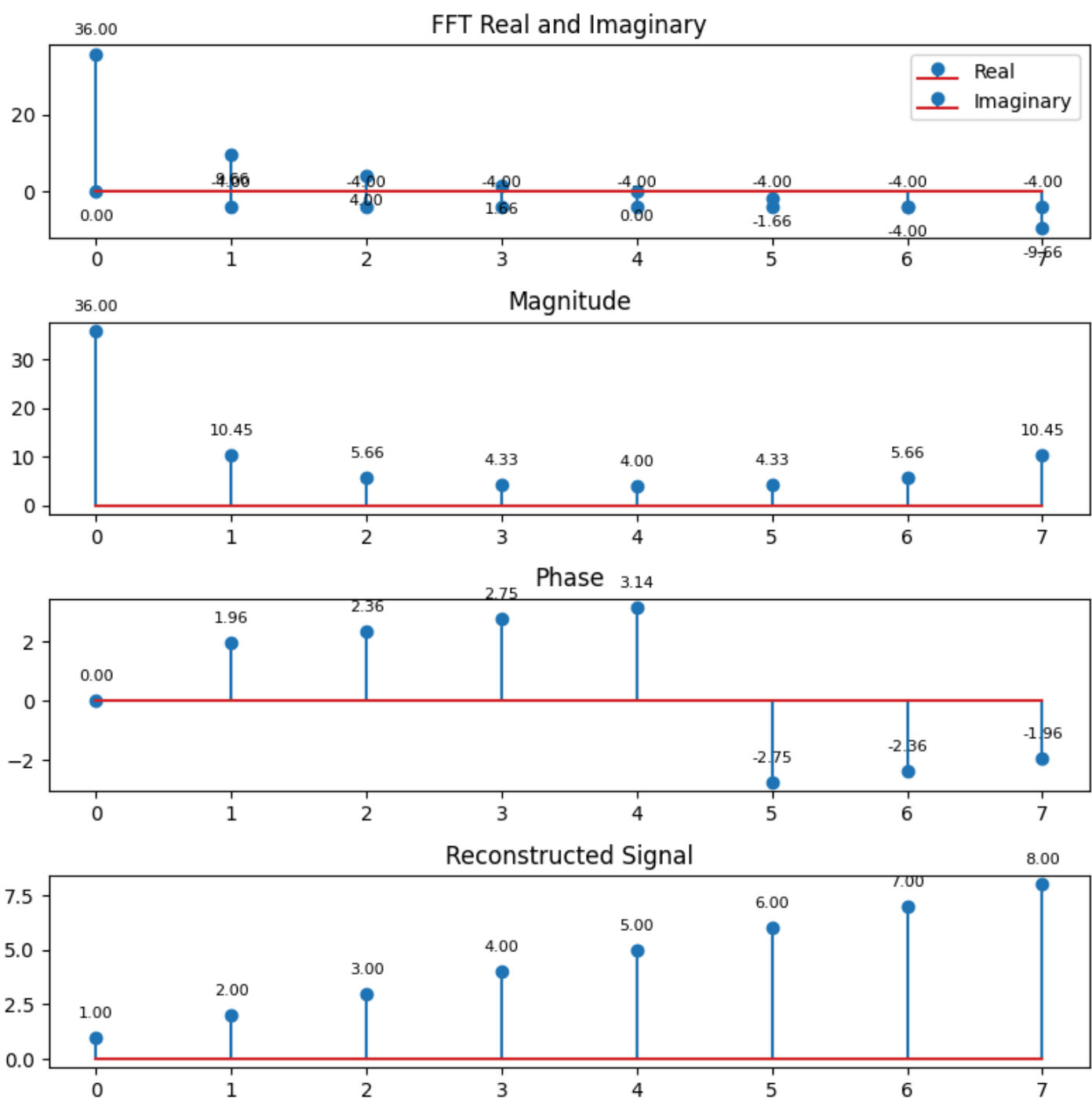
```
plt.annotate(f'{val:.2f}', (i, val), textcoords="offset points", xytext=(0,10), ha='center',
fontSize=8)
```

```
plt.title("Reconstructed Signal")
```

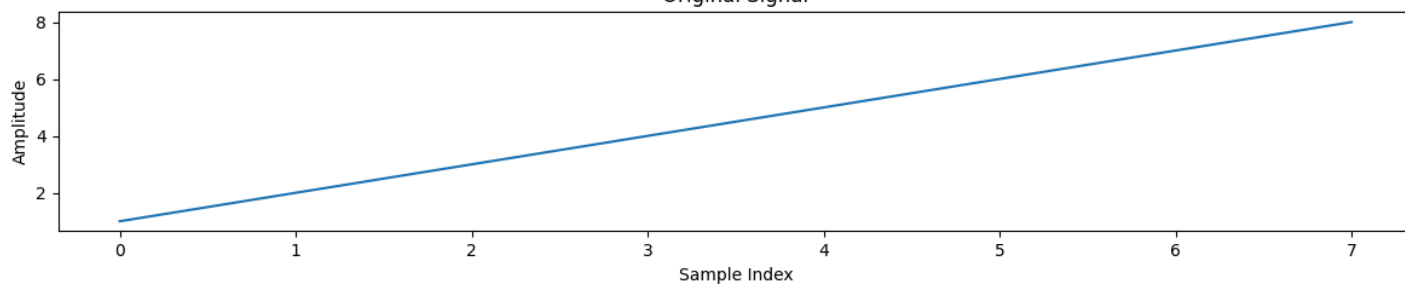
```
plt.tight_layout()
```

```
plt.show()
```

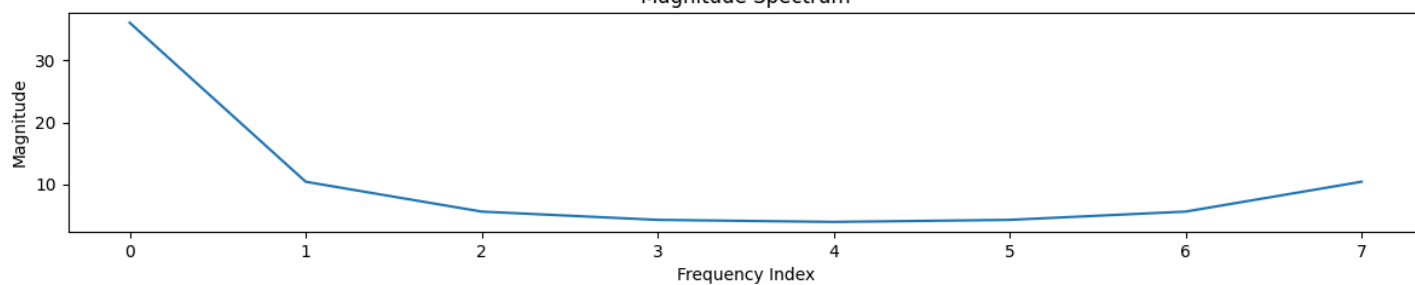
OutPut:-



Original Signal



Magnitude Spectrum



Phase Spectrum

